

Hanan Ibrahim

+94 758 520 552 — m7hanan@gmail.com — linkedin.com/in/Hanan — github.com/hanan — Portfolio

Summary — Aspiring AI & Robotics Engineer specializing in high-performance Computer Vision, Edge AI, and autonomous systems. Hands-on experience developing low-latency inference pipelines on NVIDIA Jetson hardware and deploying accelerated models (YOLO, TensorRT). Proficient in C++ and Python, with a strong understanding of hardware-software co-design, real-time data streaming, and Linux systems. Seeking to contribute to production-grade Edge AI platforms.

Skills

- **Languages:** C++, Python, C, Java, SQL
- **Edge AI & Vision:** NVIDIA Jetson Nano/Orin, TensorRT, YOLO, OpenCV, PyTorch, TensorFlow
- **Developer Tools:** Linux/Ubuntu, Git, GitHub, Docker, REST APIs, ROS 2, Gazebo, AirSim
- **Embedded & Hardware:** Raspberry Pi, NVIDIA Jetson, ESP32, STM32, Embedded C, System Monitoring

Experience

Centre for Defense Research and Development

Research Assistant (Internship)

Jan 2026 – Present

Colombo, Western Province, Sri Lanka

- Conducted research and development on AI-enabled surveillance and threat detection systems, focusing on deep learning and computer vision for aerial and RF-based monitoring.
- Contributed to projects involving intelligent detection, signal analysis, and decision-support systems for defense and security applications.
- Supported advanced research at the intersection of artificial intelligence, radio-frequency systems, and electronic warfare technologies.

Hanan Bin Ibrahim – YouTube Channel

Technical Educator & Project Developer

Oct 2019 – Present

Colombo, Sri Lanka

- Created and managed a technical YouTube channel producing AI, robotics, and drone tutorials with AirSim simulations, computer vision architecture, and embedded systems projects.

Education

University of Colombo

Bachelor of Engineering Technology Honours in Instrumentation and Automation

Dec 2023 – Expected Dec 2027

Colombo, Sri Lanka

Macan Makar National School

G.C.E. Advanced Level

Feb 2020 – Dec 2022

Batticaloa District, Sri Lanka

Projects

Featured Edge AI Project

EdgeForge-Vision: Accelerated Industrial Inspection Framework

2026

- Engineered an open-source Edge AI framework optimized for low-latency industrial inspection and telemetry on resource-constrained NVIDIA Jetson platforms.
- Implemented model quantization and optimization pipelines, converting standard deep learning weights to TensorRT (FP16/INT8) to increase inference speed by 40% and compute efficiency on Jetson GPU cores.
- Designed a multi-stream ingestion pipeline managing real-time video feeds with synchronized frame pre-processing to eliminate CPU-GPU memory transfer bottlenecks.
- Developed modular telemetry logging systems to track GPU temperature, VRAM utilization, and inference latency in real-time.
- **Tech Stack:** Python, C++ Wrapper, NVIDIA Jetson, TensorRT, OpenCV, CUDA-accelerated ops
- **GitHub:** EdgeForge-Vision — **Demo:** Watch on YouTube

Drone & Computer Vision Projects

YOLOv5-based Autonomous Drone Navigation in AirSim

2024

- Engineered a full-stack autonomy system integrating YOLOv5 object detection with AirSim for real-time drone navigation, collision avoidance, and autonomous flight operations.
- **Tech Stack:** Python, PyTorch, YOLOv5, OpenCV, AirSim API
- **GitHub:** YOLOv5-AirSim-Autonomy — **Demo:** Watch on YouTube

Precision Payload Drone Landing System in AirSim2024

– Designed and implemented autonomous drone navigation and precision landing systems in AirSim using modular Python architecture.

– Tech Stack: Python, AirSim API, UAV Navigation

– GitHub: [airsim-drone-path-landing](#)

AirSim Drone Simulator & Control Interface2024

– Developed a low-level drone control interface in AirSim with keyboard input and quaternion conversion for testing computer vision algorithms.

– Tech Stack: Python, AirSim API, Quaternion Mathematics

– GitHub: [AirSim-Drone-Keybaord-Controller](#)

Automation & Robotics Projects

Automatic Railway Gate Control System2023

– Co-developed an ESP32-based automated railway gate system with ultrasonic sensors, servo motors, and traffic lights for enhanced safety.

– Tech Stack: C++, ESP32, Ultrasonic Sensors, Servo Motors

– GitHub: [Auto-Railway-Gate-System](#)

Vision-Enhanced Line Following Robot with Obstacle Avoidance2023

– Designed and built an ATmega328P-based autonomous robot with IR sensor array and ultrasonic obstacle avoidance for precise line navigation.

– Tech Stack: Embedded C, ATmega328P, IR Sensors, Ultrasonic Sensor, Motor Control

– GitHub: [Line-following-Robot](#)

Machine Learning & AI Projects

Real-Time Face Recognition System2024

– Developed a deep learning model for facial recognition, trained on a custom dataset, achieving 95% accuracy in identifying the target user from real-time video input.

– Engineered a pipeline using OpenCV and dlib for face detection, alignment, and feature extraction, leveraging the Face Recognition library for classification.

– Tech Stack: Python, OpenCV, dlib, Face Recognition library

– GitHub: [face-recognition](#)

Additional Experience

EX-PACK CORRUGATED CARTONS PLCOct 2022 – Mar 2023

Mechanical HelperColombo, Western Province, Sri Lanka

– Assisted in machinery maintenance and repair while supporting senior technicians and adhering to workplace safety protocols.

Ceylon Transport BoardFeb 2022 – May 2022

Automotive TechnicianBatticaloa District, Eastern Province, Sri Lanka

– Supported technicians in automotive/mechanical repair, gaining hands-on technical skills and workplace safety experience.

Certifications

– Supervised Machine Learning: Regression and Classification

DeepLearning.AI

Certificate

Nov 2024

– Advanced Learning Algorithms

DeepLearning.AI

Certificate

Nov 2024

– Unsupervised Learning, Recommenders, Reinforcement Learning

DeepLearning.AI

Certificate

Nov 2024